RIKATHI PAL

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Research Profile

Computer Science Ph.D. student at Stony Brook University specializing in generative AI, computer vision, and medical image analysis. Develops controllable 3D diffusion models for CT synthesis and latent-space methods for interactive image generation. Research also spans topological analysis of neural networks and visual analytics for human oversight of AI. Experienced in model development, experimental evaluation, and scientific publication, with strong programming skills in Python, PyTorch, and TensorFlow.

Education

- **Stony Brook University, New York, United States** Fall 2025 – Expected 2030
  PhD in Computer Science — Advisor: Prof. Klaus Mueller **GPA: 3.87/4**
  – Research interests: biomedical ML, generative models, and medical imaging.

- **Information Technology, University of Calcutta, India** 2020 – 2024
  B.Tech in Information Technology — Advisors: Amlan Chakrabarti and Somoballi Ghoshal **CGPA: 9.36/10**
  – Research: lumbar spine tumor segmentation in T2-weighted MRI.

Research Experience

Research Assistant **August 2025 – Present**
Stony Brook University, New York, United States
Advisor: Prof. Klaus Mueller
– Developed controllable 3D diffusion models and latent-space methods for medical image generation.
– Conducted quantitative evaluations and downstream classification experiments; collaborated on manuscript preparation.

Teaching Assistant **August 2025 – Present**
Stony Brook University, New York, United States
– Supported day-to-day course operations by grading assignments, maintaining academic records, explaining technical concepts, and mentoring students.
– Communicated technical concepts clearly to diverse audiences through one-on-one support and instructional assistance.

Pre Doctoral Research Fellow **June 2024 – July 2025**
Indian Institute of Science (IISc), Bangalore, India
Studying Neural Network Training via Loss Landscape Analysis Using Topological Data Analysis
– Analyzed neural network training through topological data analysis of loss landscapes and latent representations.
– Applied contour trees and persistent homology to characterize topological structure.
– Collaborated with faculty on experimental evaluation, result interpretation, and research presentations.

Research Intern **May 2023 – Nov 2023**
Indian Statistical Institute (ISI), Kolkata, India
ACEV: Unsupervised Segmentation of Intersecting Manifolds in High-Dimensional Spaces
– Contributed to unsupervised manifold segmentation using k-nearest neighbors and principal component analysis (PCA).
– Investigated adaptive segmentation thresholds based on angular changes of local eigenvectors.
– Conducted experimental evaluations and presented methodological findings to research collaborators.

Research Projects

Manifold Analysis of Real and Synthetic Medical Images ***March 2026–Present***
Stony Brook University, United States — Advisor: Prof. Klaus Mueller
– Investigating the geometric and topological structure of latent manifolds learned from real and synthetic medical images.
– Combining manifold learning, dimensionality reduction, and visualization to analyze representation quality and generative behavior.
– Exploring applications to realism assessment, model interpretability, and controllable medical image generation.

AI Scientist Mission Control (AIMC) ***2026–Present***
Stony Brook University, United States — Advisor: Prof. Klaus Mueller
– Developing a visual analytics framework for human oversight of autonomous scientific discovery.
– Integrating semantic embeddings, automated weakness extraction, and temporal analysis with interactive visualizations.

– Supporting assessment of research quality, identification of recurring failure modes, and prioritization of promising discoveries for human review.

Medical Image Generation from 2D Empty Space ***2026–Present***
Stony Brook University, United States — Advisor: Prof. Klaus Mueller

– Generated CT images from user-selected empty regions in 2D projections of a high-dimensional latent space.
– Reconstructed latent vectors using distance-weighted nearest neighbors and cluster-aware selection.
– Evaluated image quality, perceptual similarity, and latent reconstruction accuracy.

Pathology-Controlled Medical Image Generation ***2025–Present***
Stony Brook University, United States — Advisor: Prof. Klaus Mueller

– Developed mask-conditioned 3D DDPMs for anatomy-aware, pathology-controlled lung CT synthesis.
– Designed phase-decoupled global and local guidance to preserve anatomy and control lesion appearance.
– Evaluated synthetic image quality and augmentation for downstream lung lesion classification.

Lumbar Spine Tumor Segmentation & Localization (T2 MRI) ***2023–2024***
B.Tech Research — University of Calcutta, India — Advisor: Amlan Chakrabarti and Somoballi Ghoshal

– Worked on deep learning methods for lumbar spine tumor segmentation and localization in T2-weighted MRI.
– Performed medical image preprocessing and experimental evaluation of segmentation results.
– Co-authored a research paper accepted at NCVPRIPG 2025.

Leaf Disease Detection Using MobileResNet ***2023–2024***
B.Tech Research — University of Calcutta, India — Advisors: Amlan Chakrabarti and Somoballi Ghoshal

– Designed MobileResNet for leaf disease detection on low-compute devices.
– Curated image datasets and performed preprocessing and feature extraction.
– Trained and evaluated the model and contributed to research documentation.

Publications

- **AI Scientist Mission Control (AIMC): Visual Analytics for Human Oversight of Autonomous Scientific Discovery** *DOI No.*
  *Authors: **Rikathi Pal***, Klaus Mueller*
  Conference: IEEE VIS 2026 Poster, Date: November 9-13, 2026

- **Medical Image Segmentation Using Advanced Attention UNet** **(1st Runner-up)** *DOI No.*
  *Authors: **Rikathi Pal***, Somoballi Ghoshal, Amlan Chakrabarti*
  Conference: ICDMAI 2025, Date: January 17-19, 2025

- **Lumbar Spine Tumour Segmentation and Localization from T2 MRI Images Using AI** *DOI No.*
  *Authors: **Rikathi Pal**, Somoballi Ghoshal, Piya Saha, Amlan Chakrabarti, Susmita Sur-Kolay*
  Conference: NCVPRIPG 2025, Date: July 19-22, 2025

- **MobileResNet: A Generalized Lightweight Model for Accurate Leaf Disease Detection Across Diverse Plant Types** *DOI No.*
  *Authors: **Rikathi Pal***, Somoballi Ghoshal, Amlan Chakrabarti*
  Conference: INDICON'24, Date: December 19-21, 2024

- **A Novel Feature Extraction Model for the Detection of Plant Disease from Leaf Images in Low Computational Devices** *DOI No.*
  *Authors: **Rikathi Pal***, Anik Basu Bhaumik, Arpan Murmu, Sanoar Hossain, Biswajit Maity, Soumya Sen*
  Conference: ICSTA'23, Date: December 19-20, 2023

- ***Under Review*** **- Anatomy-Aware and Pathology-Controlled Diffusion for 3D Lung CT Synthesis**
  *Authors: **Rikathi Pal**, Agransh Srivastava, Arjun Krishna, Klaus Mueller*
  Journal: Submitted to Physics in Medicine and Biology

- ***Under Review*** **- Topological Analysis of DNNs via Loss-Augmented Latent Spaces**
  *Authors: **Rikathi Pal**, Vijay Natarajan, Harish Doraiswamy*
  Journal: Submitted to IEEE Transactions on Visualization and Computer Graphics

Scholarship & Award

– **Hagan International Scholarship** — Hagan Scholarship Foundation (2025–2027)

– **Second Best Paper Award** — International Conference on Data Management, Analytics & Innovation (ICDMAI 2025)
Paper: *Medical Image Segmentation Using Advanced Attention UNet*

Leadership & Volunteer Experience

Student Volunteer (Upcoming) **November 2026**
IEEE VIS 2026, Boston, MA, United States

– Will support conference operations as a student volunteer.

Student Lead – AI/ML Initiatives **April 2023 – March 2024**
Google Developer Student Club (GDSC), University of Calcutta, India

– Led AI/ML team activities, coordinated technical sessions, and tracked collaborative project progress.
– Communicated project methods and outcomes through technical documentation and presentations.

Student Volunteer **January 2024**
VLSID Conference 2024, Kolkata, India

– Supported conference operations as a student volunteer.

Technical Skills

- **Programming:** Python, C++, SQL, Java.
- **Deep Learning Frameworks:** PyTorch, TensorFlow, Keras.
- **Research Methods:** Generative modeling, diffusion models, convolutional neural networks, medical image segmentation, representation learning, unsupervised learning, dimensionality reduction.
- **Scientific Computing and Computer Vision:** NumPy, Pandas, OpenCV, Matplotlib, Seaborn.
- **Development Tools:** Git, Visual Studio Code, LaTeX.

CERTIFICATIONS

- Selected as a Fellow and participated in the VLSID 2024 Conference — *Certificate*
- 7th CVIT Summer School on Computer Vision and Machine Learning, IIIT Hyderabad — *Certificate*
- Secured 3rd place in the Smart Bengal Hackathon 2024 — *Certificate*
- Computer Vision and Image Processing Essentials (IBM) — *Certificate*
- Linear Algebra for Machine Learning (Coursera) — *Certificate*
- Hands-on Introduction to IoT with Machine Learning (IIT Kharagpur) — *Certificate*
- Python for Data Science and Machine Learning (Udemy) — *Certificate*
- Neural Networks and Deep Learning (Coursera) — *Certificate*
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Coursera) — *Certificate*
- Linear Algebra for Computer Science & Machine Learning (NPTEL) — *Certificate*